

江苏省专转本 2024 届高等数学真题试卷

答案解析

1. 选 A

解：当 $x \rightarrow 0$ 时， $\alpha(x) = \cos x - 1 \sim -\frac{1}{2}x^2$

$$\beta(x) = \sqrt{1+x^2} - 1 \sim \frac{1}{2}x^2$$

$$\gamma(x) = e^x - 1 \sim x$$

$\therefore$ 当 $x \rightarrow 0$ 时， $\alpha(x)$ 是 $\beta(x)$ 的同阶无穷小， $\beta(x)$ 是 $\gamma(x)$ 的高阶无穷小

2. 选 A

解： $\because$ 极限值是一个常数， $\therefore$ 可设 $\lim_{x \rightarrow 0} f(x) = A$

$$\text{则 } f(x) = \frac{\sin 2x}{x} + 2A, \text{ 故 } \lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{\sin 2x}{x} + 2A$$

$$\text{即 } A = \lim_{x \rightarrow 0} \frac{\sin 2x}{x} + 2A, \therefore -A = \lim_{x \rightarrow 0} \frac{\sin 2x}{x} = 2, A = -2$$

$$\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} \left(\frac{\sin 2x}{x} - 4 \right) = -4$$

3. 选 D

解：因 e^{-2x} 是 $f(x)$ 的一个原函数，所以 $f(x) = (e^{-2x})' = -2e^{-2x}$

$$f'(x) = 4e^{-2x}, f''(x) = -8e^{-2x}$$

4. 选 C

$$\text{解： } f'(x) = \frac{1}{2x+1} \cdot 2, f''(x) = \frac{-1}{(2x+1)^2} \cdot 2^2$$

$$f'''(x) = \frac{1 \cdot 2}{(2x+1)^3} \cdot 2^3, f^{(4)}(x) = \frac{-1 \cdot 2 \cdot 3}{(2x+1)^4} \cdot 2^4$$

$$\therefore f^{(n)}(x) = \frac{(-1)^{n-1} \cdot 2^n \cdot (n-1)!}{(2x+1)^n}$$

5. 选 D

解: 当 $n \rightarrow \infty$ 时, $\sin \frac{1}{n} \sim \frac{1}{n}$, $\therefore \sum_{n=1}^{\infty} (-1)^n \sin \frac{1}{n}$ 收敛

6. 选 B

解: 令 $f'_x(x, y) = 6x^2 - 6x = 0$, 得 $x_1 = 0, x_2 = 1$

$$f'_y(x, y) = 2y - 2 = 0, \text{ 得 } y = 1$$

$\therefore$ 驻点为 $(0, 1), (1, 1)$

$$f''_{xx}(x, y) = 12x - 6, \quad f''_{xy}(x, y) = 0, \quad f''_{yy}(x, y) = 2$$

$$\text{在 } (0, 1) \text{ 处 } A = f''_{xx}(0, 1) = -6, \quad B = f''_{xy}(0, 1) = 0, \quad C = f''_{yy}(0, 1) = 2$$

$$\therefore AC - B^2 = -12 < 0, \therefore f(x, y) \text{ 在 } (0, 1) \text{ 处不取得极值}$$

$$\text{在 } (1, 1) \text{ 处 } A = f''_{xx}(1, 1) = 6, \quad B = f''_{xy}(1, 1) = 0, \quad C = f''_{yy}(1, 1) = 2$$

$$\therefore AC - B^2 = 12 > 0, \therefore f(x, y) \text{ 在 } (1, 1) \text{ 处取得极小值}$$

7. 选 D

$$\text{解: } \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & -1 & 2 & -2 & 3 \\ 1 & 1 & 4 & 4 & 9 \\ 1 & -1 & 8 & -8 & 27 \end{pmatrix} \xrightarrow{\substack{r_2-r_1 \\ r_3-r_1 \\ r_4-r_1}} \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & -2 & 1 & -3 & 2 \\ 0 & 0 & 3 & 3 & 8 \\ 0 & -2 & 7 & -9 & 26 \end{pmatrix}$$

$$\xrightarrow{r_4-r_2} \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & -2 & 1 & -3 & 2 \\ 0 & 0 & 3 & 3 & 8 \\ 0 & 0 & 6 & -6 & 24 \end{pmatrix} \xrightarrow{r_4-2r_3} \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & -2 & 1 & -3 & 2 \\ 0 & 0 & 3 & 3 & 8 \\ 0 & 0 & 0 & -12 & 8 \end{pmatrix}$$

所以矩阵的秩为 4

8. 选 B

$$\text{解: 对于 B } (\alpha_1 - \alpha_2, \alpha_2 - \alpha_3, \alpha_3 - \alpha_1) = (\alpha_1, \alpha_2, \alpha_3) \begin{pmatrix} 1 & 0 & -1 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix}$$

$$\therefore |\alpha_1 - \alpha_2, \alpha_2 - \alpha_3, \alpha_3 - \alpha_1| = |\alpha_1, \alpha_2, \alpha_3| \begin{vmatrix} 1 & 0 & -1 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{vmatrix} = 0$$

9. 解: $\because x=1$ 是 $f(x) = \frac{x^3 - ax}{x^2 - x}$ 的第一类间断点

$$\therefore \lim_{x \rightarrow 1} f(x) = \lim_{x \rightarrow 1} \frac{x^3 - ax}{x^2 - x} = \lim_{x \rightarrow 1} \frac{x^2 - a}{x - 1} \text{ 存在}$$

$$\because \lim_{x \rightarrow 1} (x - 1) = 0, \therefore \text{必有} \lim_{x \rightarrow 1} (x^2 - a) = 1 - a = 0, \text{ 故 } a = 1$$

$$\text{于是} \lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{x^3 - ax}{x^2 - x} = \lim_{x \rightarrow 0} \frac{x^2 - a}{x - 1} = a = 1$$

10. 解: $\frac{dx}{dt} = 2t + 2, \quad \frac{dy}{dt} = 3t^2 - 3$

$$\frac{dy}{dx} = \frac{dy}{dt} \bigg/ \frac{dx}{dt} = \frac{3t^2 - 3}{2t + 2}$$

$$\text{令} \frac{dy}{dx} = -\frac{3}{2}, \text{ 即 } \frac{3t^2 - 3}{2t + 2} = -\frac{3}{2}, \text{ 解得 } t = 0$$

11. 解: $y = f(\sin x) = \begin{cases} \frac{\ln(1 + \sin^2 x)}{\sin x} & x \neq 0 \\ 0 & x = 0 \end{cases}$

$$\left. \frac{dy}{dx} \right|_{x=0} = f'(0) = \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x} = \lim_{x \rightarrow 0} \frac{\frac{\ln(1 + \sin^2 x)}{\sin x} - 0}{x}$$

$$= \lim_{x \rightarrow 0} \frac{\ln(1 + \sin^2 x)}{x \sin x} = \lim_{x \rightarrow 0} \frac{\sin^2 x}{x \sin x} = 1$$

12. 解: $\int_{-\infty}^a e^x dx = \lim_{b \rightarrow -\infty} \int_b^a e^x dx = \lim_{b \rightarrow -\infty} e^x \bigg|_b^a = \lim_{b \rightarrow -\infty} (e^a - e^b) = e^a$

$$\begin{aligned} \int_a^{+\infty} e^{1-x} dx &= \lim_{c \rightarrow +\infty} \int_a^c e^{1-x} dx = - \lim_{c \rightarrow +\infty} \int_a^c e^{1-x} d(1-x) = - \lim_{c \rightarrow +\infty} e^{1-x} \bigg|_a^c \\ &= - \lim_{c \rightarrow +\infty} (e^{1-c} - e^{1-a}) = e^{1-a} \end{aligned}$$

$$\text{由 } e^a = e^{1-a}, \text{ 得 } a = 1 - a, \text{ 故 } a = \frac{1}{2}$$

$$13. \text{ 解: } \lim_{n \rightarrow \infty} \left| \frac{(n+1)^{n+1} (x-1)^{n+1}}{3^{n+1} \cdot (n+1)!} \cdot \frac{3^n \cdot n!}{n^n (x-1)^n} \right| = \lim_{n \rightarrow \infty} \left| \frac{(n+1)^n}{3 \cdot n^n} \right| |x-1|$$

$$= \lim_{n \rightarrow \infty} \frac{1}{3} \left(1 + \frac{1}{n} \right)^n |x-1| = \frac{e}{3} |x-1| < 1$$

$$\therefore |x-1| < \frac{3}{e}, \text{ 收敛半径 } R = \frac{3}{e}$$

$$14. \text{ 解: } \begin{vmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 2 & 0 \\ 0 & 3 & 4 & 0 \\ 3 & 0 & 0 & 4 \end{vmatrix} \xrightarrow{r_4 - 3r_1} \begin{vmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 2 & 0 \\ 0 & 3 & 4 & 0 \\ 0 & 0 & 0 & -2 \end{vmatrix} = \begin{vmatrix} 1 & 2 & 0 \\ 3 & 4 & 0 \\ 0 & 0 & -2 \end{vmatrix} = -2 \begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix} = 4$$

$$15. \text{ 解: } \text{令 } x = \frac{1}{t}, \text{ 当 } x \rightarrow \infty \text{ 时 } t \rightarrow 0$$

$$\text{原式} = \lim_{t \rightarrow 0} \frac{\arctan \frac{1}{t^2} - \frac{\pi}{2}}{t^2} = \lim_{t \rightarrow 0} \frac{\frac{1}{1 + \frac{1}{t^4}} \cdot \frac{-2}{t^3}}{2t} = \lim_{t \rightarrow 0} \frac{\frac{-2t}{1+t^4}}{2t} = -\lim_{t \rightarrow 0} \frac{1}{1+t^4} = -1$$

$$16. \text{ 解: } \text{令 } \sqrt{x+2} = t, \text{ 则 } x = t^2 - 2$$

$$\begin{aligned} \text{原式} &= \int \frac{t-1}{t(t^2+1)} d(t^2-2) = 2 \int \frac{t-1}{t^2+1} dt \\ &= 2 \int \frac{t}{t^2+1} dt - 2 \int \frac{1}{t^2+1} dt = \ln(1+t^2) - 2 \arctan t + c \\ &= \ln|x+3| - 2 \arctan \sqrt{x+2} + c \end{aligned}$$

$$17. \text{ 解: } \int_0^1 \frac{x}{x+1} \sqrt{1-x^2} dx = \int_0^1 \sqrt{1-x^2} dx - \int_0^1 \frac{\sqrt{1-x^2}}{x+1} dx = \frac{\pi}{4} - \int_0^1 \frac{\sqrt{1-x^2}}{x+1} dx$$

$$\text{对于 } \int_0^1 \frac{\sqrt{1-x^2}}{x+1} dx, \text{ 令 } x = \sin t, \text{ 当 } x=0 \text{ 时 } t=0, \text{ 当 } x=1 \text{ 时 } t = \frac{\pi}{2}$$

$$\text{所以 } \int_0^1 \frac{\sqrt{1-x^2}}{x+1} dx = \int_0^{\frac{\pi}{2}} \frac{\sqrt{1-\sin^2 t}}{\sin t + 1} d \sin t = \int_0^{\frac{\pi}{2}} \frac{\cos^2 t}{\sin t + 1} dt$$

$$= \int_0^{\frac{\pi}{2}} \frac{1 - \sin^2 t}{\sin t + 1} dt = \int_0^{\frac{\pi}{2}} (1 - \sin t) dt$$

$$= (t + \cos t) \Big|_0^{\frac{\pi}{2}} = \frac{\pi}{2} - 1$$

$$\text{原式} = \frac{\pi}{4} - \frac{\pi}{2} + 1 = 1 - \frac{\pi}{4}$$

18. 解: 设所求微分方程为 $y'' + py' + qy = f(x)$

$\because y_1 = e^{3x}, y_2 = e^x + e^{3x}, y_3 = e^{2x} + e^{3x}$ 是它得三个特解

$\therefore y_2 - y_1 = e^x, y_3 - y_1 = e^{2x}$ 是对应齐次方程的解

$\therefore$ 对应齐次方程的通解为 $\bar{y} = c_1 e^x + c_2 e^{2x}$

特征根 $r_1 = 1, r_2 = 2$, 特征方程 $(r-1)(r-2) = 0$, 即 $r^2 - 3r + 2 = 0$

$\therefore$ 对应齐次方程为 $y'' - 3y' + 2y = 0$

又 $\because y = e^{3x}$ 是 $y'' - 3y' + 2y = f(x)$ 的解

$\therefore (e^{3x})'' - 3(e^{3x})' + 2(e^{3x}) = f(x)$, 解得 $f(x) = 2e^{3x}$

所求微分方程为 $y'' - 3y' + 2y = 2e^{3x}$

19. 解: 方程两边对 x 求偏导数, 得:

$$\frac{1}{1+(x+2y+3z)^2} \left(1+3\frac{\partial z}{\partial x} \right) - yz - xy \frac{\partial z}{\partial x} = 0 \quad \textcircled{1}$$

方程两边对 y 求偏导数, 得:

$$\frac{1}{1+(x+2y+3z)^2} \left(2+3\frac{\partial z}{\partial y} \right) - xz - xy \frac{\partial z}{\partial y} = 0 \quad \textcircled{2}$$

由原方程可知, 当 $x=0, y=0$ 时, $\arctan 3z = 0$, $\therefore z = 0$

将 $x=0, y=0, z=0$ 带入①式, 得: $1+3\frac{\partial z}{\partial x} = 0$, $\therefore \frac{\partial z}{\partial x} = -\frac{1}{3}$

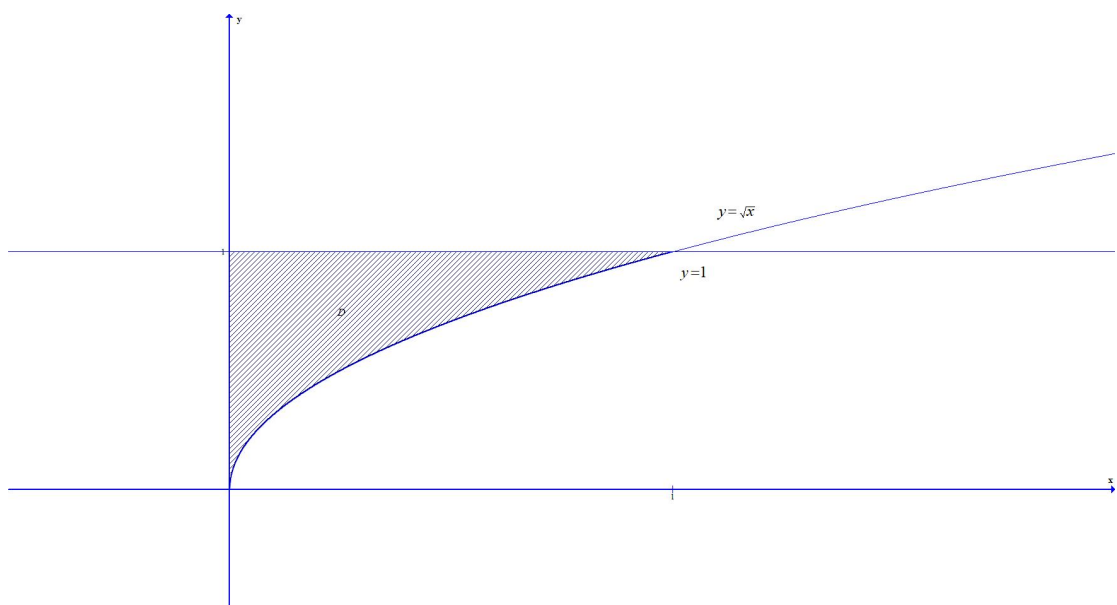
将 $x=0, y=0, z=0$ 带入②式, 得: $2+3\frac{\partial z}{\partial y} = 0$, $\therefore \frac{\partial z}{\partial y} = -\frac{2}{3}$

$$\therefore dz|_{(0,0)} = \frac{\partial z}{\partial x}|_{(0,0)} dx + \frac{\partial z}{\partial y}|_{(0,0)} dy = -\frac{1}{3}dx - \frac{2}{3}dy$$

20. 解: 积分区域 $D: \begin{cases} \sqrt{x} \leq y \leq 1 \\ 0 \leq x \leq 1 \end{cases}$

交换积分次序得:

$$\begin{aligned}
 \int_0^1 dx \int_{\sqrt{x}}^1 \frac{\cos y - 1}{y} dy &= \int_0^1 dy \int_0^{y^2} \frac{\cos y - 1}{y} dx = \int_0^1 dy \cdot \frac{\cos y - 1}{y} \cdot x \Big|_0^{y^2} \\
 &= \int_0^1 y (\cos y - 1) dy = \int_0^1 y \cos y dy - \int_0^1 y dy \\
 &= \int_0^1 y d \sin y - \frac{1}{2} y^2 \Big|_0^1 = y \sin y \Big|_0^1 - \int_0^1 \sin y dy - \frac{1}{2} \\
 &= \sin 1 + \cos y \Big|_0^1 - \frac{1}{2} = \sin 1 + \cos 1 - \frac{3}{2}
 \end{aligned}$$



$$21. \text{ 解: } (A:E) = \begin{pmatrix} 1 & 1 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{pmatrix} \xrightarrow{r_2 - r_1} \begin{pmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & -1 & 1 \end{pmatrix} \xrightarrow{r_1 - r_2} \begin{pmatrix} 1 & 0 & 2 & -1 \\ 0 & 1 & -1 & 1 \end{pmatrix}$$

$$\therefore A^{-1} = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}$$

$$(B:E) = \begin{pmatrix} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{pmatrix} \xrightarrow{r_1 - r_2} \begin{pmatrix} 1 & 0 & 0 & 1 & -1 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{pmatrix}$$

$$\xrightarrow{r_2 - r_3} \begin{pmatrix} 1 & 0 & 0 & 1 & -1 & 0 \\ 0 & 1 & 0 & 0 & 1 & -1 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{pmatrix}$$

$$\therefore B^{-1} = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}$$

$$X = A^{-1}CB^{-1} = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 3 & 4 \\ 1 & 4 & 5 \end{pmatrix} \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$22. \text{ 解: } \bar{A} = \begin{pmatrix} 1 & 1 & -2 & 0 & 5 \\ 2 & 1 & -1 & 1 & 7 \\ 1 & 2 & -5 & -1 & 8 \end{pmatrix} \xrightarrow[r_3-r_1]{r_2-2r_1} \begin{pmatrix} 1 & 1 & -2 & 0 & 5 \\ 0 & -1 & 3 & 1 & -3 \\ 0 & 1 & -3 & -1 & 3 \end{pmatrix}$$

$$\xrightarrow[r_2]{r_3+r_2} \begin{pmatrix} 1 & 1 & -2 & 0 & 5 \\ 0 & 1 & -3 & -1 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{r_1-r_2} \begin{pmatrix} 1 & 0 & 1 & 1 & 2 \\ 0 & 1 & -3 & -1 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

x_1, x_2 为约束变量, x_3, x_4 为自由变量

$$\text{一般解} \begin{cases} x_1 = 2 - x_3 - x_4 \\ x_2 = 3 + 3x_3 + x_4 \end{cases}, \text{ 令 } x_3 = x_4 = 0 \text{ 得原方程组一个特解 } \eta^* = \begin{pmatrix} 2 \\ 3 \\ 0 \\ 0 \end{pmatrix}$$

$$\text{导出组一般解} \begin{cases} x_1 = -x_3 - x_4 \\ x_2 = 3x_3 + x_4 \end{cases}$$

$$\text{令} \begin{cases} x_3 = 1 \\ x_4 = 0 \end{cases}, \begin{cases} x_3 = 0 \\ x_4 = 1 \end{cases}, \text{ 得基础解系 } \xi_1 = \begin{pmatrix} -1 \\ 3 \\ 1 \\ 0 \end{pmatrix}, \xi_2 = \begin{pmatrix} -1 \\ 1 \\ 0 \\ 1 \end{pmatrix}$$

$$\text{方程组通解为} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \\ 0 \\ 0 \end{pmatrix} + k_1 \begin{pmatrix} -1 \\ 3 \\ 1 \\ 0 \end{pmatrix} + k_2 \begin{pmatrix} -1 \\ 1 \\ 0 \\ 1 \end{pmatrix} \quad (\text{其中 } k_1, k_2 \text{ 为任意常数})$$

23. 证明:

①令 $F(x) = f(x) - x$, 则 $F(x)$ 在 $[0, 1]$ 上连续

$$\text{且 } F(0) = f(0) = 1 > 0, \quad F(1) = f(1) - 1 = -1 < 0$$

由零点定理可知, 至少存在一点 $\eta \in (0, 1)$, 使得 $F(\eta) = 0$

$$\text{即 } f(\eta) - \eta = 0, \text{ 所以 } f(\eta) = \eta$$

②令 $F(x) = xf(x) - x^2$, 则 $F(x)$ 在 $[0, \eta]$ 上连续, 在 $(0, \eta)$ 内可导

且 $F(0)=0$, $F(\eta)=\eta f(\eta)-\eta^2=\eta^2-\eta^2=0$ (由①证得到 $f(\eta)=\eta$)

由罗尔定理可知, 至少存在一点 $\xi \in (0,\eta) \subset (0,1)$, 使得 $F'(\xi)=0$

而 $F'(x)=f(x)+xf'(x)-2x$, $F'(\xi)=0$ 即为 $\xi f'(\xi)+f(\xi)=2\xi$

24. 解: ① $f'(x)-f(x)=e^2(2x-4)$ 的通解为

$$\begin{aligned} f(x) &= e^{\int dx} \left[\int e^x (2x-4) e^{-\int dx} dx + c \right] \\ &= e^x \left[\int e^x (2x-4) e^{-x} dx + c \right] \\ &= e^x \left[\int (2x-4) dx + c \right] \\ &= e^x (x^2 - 4x + c) \end{aligned}$$

由 $f(0)=5$ 可知, $c=5$

$$\therefore f(x) = (x^2 - 4x + 5)e^x$$

② $f(x)$ 的定义域为 $(-\infty, +\infty)$

$$f'(x) = (2x-4)e^x + (x^2 - 4x + 5)e^x$$

$$f''(x) = 2e^x + (2x-4)e^x + (2x-4)e^x + (x^2 - 4x + 5)e^x = (x^2 - 1)e^x$$

令 $f''(x)=0$, 得 $x=-1, x=1$

x	$(-\infty, -1)$	-1	$(-1, 1)$	1	$(1, +\infty)$
$f''(x)$	+	0	-	0	+
$y=f(x)$	$\cup$	拐点	$\cap$	拐点	$\cup$

由表可知, 曲线 $y=f(x)$ 的凹区间为 $(-\infty, -1)$, $(1, +\infty)$, 凸区间为 $(-1, 1)$

$$\text{拐点} \left(-1, \frac{10}{e} \right), (1, 2e)$$

25. 解: $S_{D_t} = \int_0^t f(x) dx$, 由题意有 $\int_0^t f(x) dx = t \ln t - t + 1$

两边对 t 求导, 得: $f(t) = \ln t$, 当 $t=e$ 时

$$\begin{aligned}
 V_x &= \pi \int_1^e (\ln t)^2 dt = \pi \int_1^e \ln^2 t dt = \pi \left[t \ln^2 t \Big|_1^e - \int_1^e t d \ln^2 t \right] \\
 &= \pi e - \pi \int_1^e \frac{t \cdot 2 \ln t}{t} dt = \pi e - 2\pi \int_1^e \ln t dt \\
 &= \pi e - 2\pi \left[t \ln t \Big|_1^e - \int_1^e t d \ln t \right] = \pi e - 2\pi \left(e - \int_1^e dt \right) \\
 &= \pi e - 2\pi e + 2\pi t \Big|_1^e = \pi e - 2\pi
 \end{aligned}$$

